

DARSHAN THAKER

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RESEARCH INTERESTS

I develop inference-time algorithms that steer pretrained generative models towards desired objectives. My work spans three directions:

- **Language Model Steering:** Efficient reward-model guidance and activation-space steering.
- **Diffusion Model Steering:** Posterior sampling algorithms for image restoration and medical imaging.
- **LLM Robustness:** Optimization-based hallucination elicitation and jailbreak attacks.

EDUCATION

University of Pennsylvania

Sep 2020 – Dec 2026 (expected)

Ph.D., Computer Science.

Columbia University

2018 – 2019

M.S., Computer Science.

The University of Texas at Austin

2014 – 2018

B.S., Computer Science and Pure Mathematics.

EXPERIENCE

University of Pennsylvania

Sep 2020 – Present

Ph.D. Researcher. Advisor: Prof. René Vidal.

Inference-Time Alignment and Steering of Language Models ([Preprint 2026](#); [NeurIPS 2024](#))

- Developed FastGuide, a decoding algorithm that makes diffusion LLM guidance with reward models up to 4.4 $\times$ faster while maintaining generation quality, using KV caching and sparse attention recomputation.
- Co-designed PaCE, an inference-time framework that uses sparse coding to identify and steer concepts in LLM activation spaces, controlling model behavior without retraining.

Red-Teaming and Hallucination Elicitation for LLMs ([ICML 2026](#); [NeurIPS 2025](#); [Preprint 2025](#))

- Mentored a junior PhD student in developing optimization-based methods (SECA, REALISTA) that elicit hallucinations from LLMs.
- Contributed to KDA, a distilled attacker that generates diverse jailbreak prompts for LLMs.

Controllable Generation with Diffusion Models ([ICCV 2025](#); [Preprint 2026](#))

- Introduced Frequency-Guided Posterior Sampling (FGPS), a training-free method that uses timestep-dependent frequency uncertainty information to steer diffusion models to restore degraded images.
- Developed ANaLOG, a probabilistic framework to model uncertainty in latent diffusion models for solving ill-posed image inverse problems.

Defenses for Adversarial Attacks in Computer Vision ([ICML 2022](#); [Preprint 2023](#); [ICLR-W 2026](#))

- Developed sparse coding algorithms that recover the deception type for adversarially perturbed images.
- Co-developed a framework extension to defend vision-language models against semantic attacks.

Siemens Healthineers

Summer 2025

Research Intern. Mentor: Dr. Mariappan Nadar.

- Trained a scale-cascaded diffusion model from scratch on a multi-GPU cluster and developed a posterior-sampling algorithm to steer it, applied to medical image super-resolution ([ISBI 2026](#), *oral*).

Salesforce Research

Spring 2020

Research Intern. Mentor: Dr. Yu Bai.

- Analyzed theoretical conditions under which deep networks learn transferable features, focusing on lower bounds on generalization error.

Earlier Industry Experience

The Curious AI Company	Research Intern, Industrial Process Control	<i>Summer 2018</i>
Facebook	Software Engineering Intern, Search Ranking	<i>Summer 2017</i>
Google	Software Engineering Intern, Location Segmentation	<i>Summer 2016</i>
Symantec	Machine Learning Intern, Center for Advanced ML	<i>Summer 2015</i>

SELECTED PUBLICATIONS

- **D. Thaker**, L. MacDonald, R. Vidal. *FastGuide: Accelerating Reward Guidance for Diffusion Large Language Models*. *Preprint 2026*
- **D. Thaker**, L. MacDonald, R. Vidal. *ANaLOG: Anisotropic Native-Latent Operator Guidance for Solving Inverse Problems*. arXiv:2609.31933 *Preprint 2026*
- B. Liang, J. Luo, L. Peng, K. Chan, **D. Thaker**, K. A. Kinfu, F. Tian, H. Hassani, R. Vidal. *REALISTA: Realistic Latent Adversarial Attacks that Elicit LLM Hallucinations*. arXiv:2605.12813 *ICML 2026*
- N. Nguyen, **D. Thaker**, K. Emmanouilidis, T. Ding, R. Vidal. *SSCD: Sparse Semantic Concept Defense Against Semantic Adversarial Attacks to Image Classifiers*. *ICLR-W 2026*
- **D. Thaker**, M. Mostapha, R. Miron, S. Qiu, M. Nadar. *Scale-Cascaded Diffusion Models for Super-Resolution in Medical Imaging*. arXiv:2601.23201 *ISBI 2026 (oral)*
- B. Liang, L. Peng, J. Luo, **D. Thaker**, K. Chan, R. Vidal. *SECA: Semantically Equivalent & Coherent Attacks for Eliciting LLM Hallucinations*. arXiv:2510.04398 *NeurIPS 2025*
- **D. Thaker**, A. Goyal, R. Vidal. *Frequency Guided Posterior Sampling for Diffusion-Based Image Restoration*. arXiv:2411.15295 *ICCV 2025*
- B. Liang, K. Chan, **D. Thaker**, J. Luo, R. Vidal. *KDA: A Knowledge-Distilled Attacker for Generating Diverse Prompts to Jailbreak LLMs*. arXiv:2502.05223 *Preprint 2025*
- J. Luo, T. Ding, K. Chan, **D. Thaker**, A. Chattopadhyay, C. Callison-Burch, R. Vidal. *PaCE: Parsimonious Concept Engineering for Large Language Models*. arXiv:2406.04331 *NeurIPS 2024*
- **D. Thaker**, P. Giampouras, R. Vidal. *A Linearly Convergent GAN Inversion-based Algorithm for Reverse Engineering of Deceptions*. arXiv:2306.04756 *CPAL 2024*
- **D. Thaker***, P. Giampouras*, R. Vidal. *Reverse Engineering ℓ_p Attacks: A Block-Sparse Optimization Approach with Recovery Guarantees*. arXiv:2203.04886 *ICML 2022*

TEACHING

Teaching Assistant, University of Pennsylvania <i>Deep Generative Models (Prof. René Vidal).</i>	<i>Fall 2023/Fall 2024</i>
Course Assistant, Columbia University <i>Deep Learning (Prof. Iddo Drori), Analysis of Algorithms (Prof. Alexandr Andoni).</i>	<i>2019</i>

HONORS AND AWARDS

ICCV Doctoral Consortium	<i>2025</i>
AWS-AI ASSET Fellowship, University of Pennsylvania	<i>2024</i>
Course Assistant Fellowship (full tuition waiver), Columbia University	<i>2019</i>
Turing Scholars Honors, The University of Texas at Austin	<i>2018</i>
University Honors, The University of Texas at Austin	<i>2014–2018</i>

SKILLS

Programming	Python, Bash, C++
ML	PyTorch, TensorFlow, Hugging Face Transformers, Diffusers
Distributed GPU Computing	PyTorch Distributed, Triton
Infrastructure	SLURM, AWS